

PHD MASTER TECHNICAL & UI SPECIFICATION

SOSA-Aligned Open-Set Radar Emitter Identification & Deinterleaving System

Full Integration Baseline | Models, Hyperparameter Control, Dataset Management & Telemetry

1. SYSTEM OVERVIEW & BATTLEFIELD REQUIREMENTS

This master specification establishes the software and hardware requirements for a mission-critical Electronic Support Measures (ESM) processing suite. Modern tactical environments present dense pulse streams exceeding 1,000,000 pulses per second, featuring agile frequency patterns, staggered PRIs, and uncataloged "zero-day" radar modes.

- **Open-Set Recognition (OSR):** Classifies known radar families with high confidence while isolating out-of-distribution (OOD) unknown signals as actionable anomalies.
- **SOSA Alignment:** Conforms strictly to Sensor Open Systems Architecture standards to minimize C-SWaP (Cost, Size, Weight, and Power) on edge tactical hardware.
- **Deterministic Real-Time Processing:** Sub-millisecond latency pipeline connecting VITA 49 hardware streams directly to automated Electronic Countermeasures (ECM).

2. PULSE DESCRIPTOR WORD (PDW) TECHNICAL SCHEMA

PDW Feature	Unit	Tactical Significance	Processing Pipeline Stage
Carrier Frequency (\$f_c\$)	MHz / GHz	Spectral band mapping & frequency agility tracking.	VITA 49 Payload Pre-filter
Pulse Width (\$PW\$)	μs	Pulse duration & modulation mode isolation.	1D-CNN Feature Extraction
ΔToA (PRI)	μs	Rhythm pattern (fixed, staggered, jittered, sliding).	Temporal Sequence Encoder
Amplitude (\$RSSI\$)	dBm	Signal power normalization & envelope bounds.	Front-end Signal Conditioning
Time of Arrival (\$ToA\$)	ns	Precision timestamp for multi-emitter deinterleaving.	FPGA Window Buffer
Pulse Rise Time (\$t_r\$)	ns	Transmitter hardware fingerprinting & emitter lineage.	Dense Metric Embedding Head

3. COMPREHENSIVE INTERACTIVE UI SPECIFICATION

3.1 Model Switcher & Architecture Control Panel

- **Neural Backbone Selector:** Dynamically toggle between **1D-CNN (Spatial Focus)**, **Transformer / LSTM (Temporal Sequence Focus)**, and **Hybrid 1D-CNN + Transformer** without restarting the backend service.
- **Distance Metric Selector:** Switch between **Cosine Similarity**, **Euclidean Distance (\$L_2\$)**, and **Dot Product** for metric space evaluation in Qdrant.
- **Inference Engine Target:** Toggle execution engine between PyTorch Native, ONNX Runtime (CPU/CUDA), and TensorRT FP16 acceleration.

3.2 Hyperparameter Tuning Studio

- **Neural Network Parameters:** Learning Rate (10^{-5} to 10^{-2}), Triplet Loss Margin ($\alpha \in [0.1, 1.0]$), Dropout Rate, and Dense Embedding Dimension (64, 128, 256).
- **Qdrant Indexing Parameters:** HNSW M (connectivity, default 16), ef_construct (search accuracy vs build speed), and Quantization Mode (Scalar SQ8 vs Binary).
- **Fusion & Safety Thresholds:** Classifier Confidence Minimum ($\text{au}_{\text{cls}} \geq 0.90$) and OOD Vector Distance Threshold (au_{dist}).

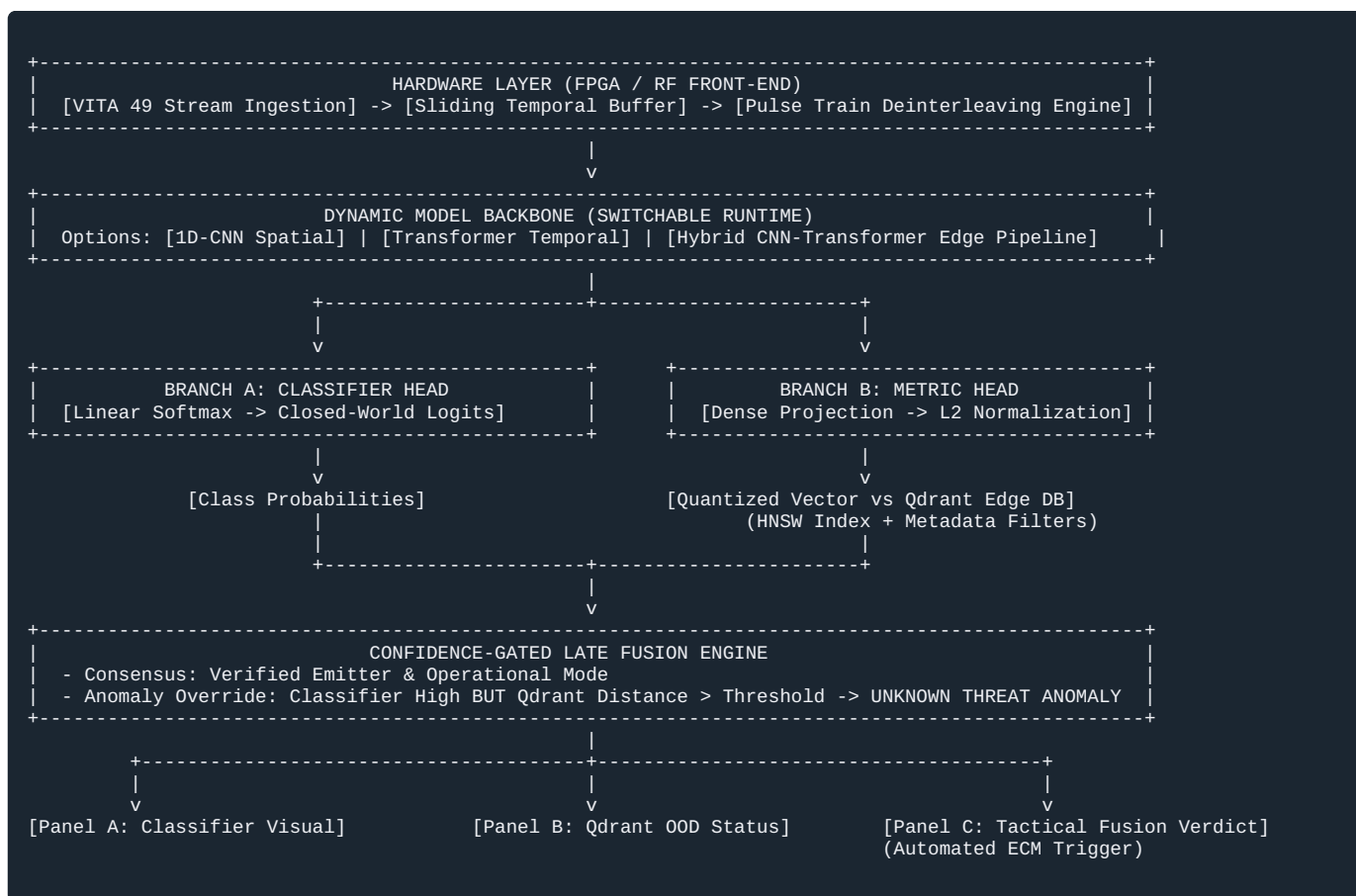
3.3 Dataset Management & Browser Suite

- **Dataset Upload Module:** Drag-and-drop interface for binary VITA 49 (`.vrt`), raw PDW CSV, and Parquet data files.
- **Synthetic Stream Generator:** Slider controls for active radar count (up to 20 families x 4 modes), noise/clutter percentage (0-30%), and pulse drop rates.
- **Dataset Inspector & Splitter:** Interactive distribution histogram of classes, operational modes, and train/validation/open-set test split partitions.

3.4 Real-Time Tri-Panel Telemetry Dashboard

- **Panel A (Closed-World Classifier Visual):** Dynamic bar chart displaying top-3 predicted emitter classes and softmax confidence scores.
- **Panel B (Qdrant OOD Metric Distance):** Real-time distance plot against known prototype centroids with highlighted anomaly rejection zones.
- **Panel C (Tactical Fusion & ECM Verdict):** High-visibility operational status badge (Verified Target Lock vs UNKNOWN THREAT ANOMALY) with automated ECM jammer trigger status.

4. SYSTEM ARCHITECTURE DIAGRAM



5. DEPLOYMENT ARCHITECTURE & VS CODE ENVIRONMENT

The system is organized into modular Python packages orchestrated locally within **VS Code** using Docker Compose for background services:

- `/config`: Dynamic YAML configuration files updated in real-time by the UI hyperparameter panel.
- `/src/data`: VITA 49 packet parsers, PDW stream loaders, and synthetic noise generators.
- `/src/models`: Switchable neural architectures (1D-CNN, Transformer, Triplet Projection Heads).
- `/src/vector_db`: Qdrant Edge client integration with Binary Quantization and metadata pre-filtering.
- `/src/api`: FastAPI server with WebSocket endpoint (`/ws/telemetry`) broadcasting sub-millisecond inference payloads.
- `/ui`: Interactive dashboard featuring model management controls, hyperparameter sliders, and real-time canvas visualizations.